

Abdullah Mohammad Aref

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Professional Summary

Versatile data scientist, backend developer and 5th-semester computer science student at Gisma University of Applied Sciences specializing in predictive modelling, machine learning algorithms, database management, and RESTful APIs. Experienced in utilizing Python, R, NumPy, and Scikit-Learn to leverage datasets and engineer insightful machine learning models and Django, Node.js, Express.js, TypeScript, MongoDB, and SQLite3 to create secure and stable backend applications. A data-driven problem solver striving to build scalable software solutions that address real-world problems and provide actionable data insights.

Education

Gisma University of Applied Sciences - BSc Computer Science Berlin, Germany 2023 – Current

- Relevant Modules: Artificial Intelligence & Machine Learning, Applied Statistical Modeling, Software Engineering, Data Structures & Algorithms, Cyber Security, App & Web Development, Databases & Big Data.

Projects

House Pricing Estimator | Python, Pandas, Scikit-Learn: 07/2025 – 09/2025

- Built an end-to-end machine learning regression analysis pipeline in Python to estimate Housing prices with high accuracy, utilizing Pandas library for transforming and cleaning data, and NumPy, Seaborn, and Matplotlib for data exploration and plotting.
- Engineered features using Scikit-Learn's ColumnTransformer with OneHotEncoder and StandardScaler to optimize numerical and categorical data efficiently
- Optimized models with hyperparameter tuning using RandomizedSearchCV and K-fold cross-validation to ensure reliable results.
- Achieved a high R2 score using Random Forest Regressor model, explaining 83.1% of factors that affect housing prices.

Telecom Customer Churn Prediction | R, Logistic Regression: 07/2025 – 09/2025

- Conducted a fully-fledged statistical analysis using R to predict customer churn, utilizing ggplot2 and ggcorrplot to explore and visualize data.
- Applied data preprocessing, sampling and cleaning with methods like scaling numerical data to reduce the effect of outliers.
- Engineered hypotheses and tested them using Chi-squared tests for categorical variables and Binomial Logistic Regression for continuous variables.
- Optimized the model results by calculating Variance Inflation Factor (VIF) to detect and remove multicollinearity, utilized ROC curve and AUC metrics to evaluate and enhance the model's predictive power by maximizing recall to 94.7% in order to effectively capture potential churning customers.

E-commerce Web Application | Vue.js, Express.js, Node.js, MongoDB, JavaScript 01/2026 – 03/2026

- Developed a full-stack e-commerce store with the MEVN stack featuring a responsive user interface with CSS Flexbox, Grid, and media queries.
- Built RESTful APIs to support CRUD operations, implementing Secure authentication, authorization, and access control using Bcrypt to hash and salt passwords.

- Integrated PayPal Sandbox API to support payments, Managed user and cart state with Pinia, and optimized image processing with Multer and Sharp
- Deployed backend applications with compiled frontend files in a Docker container and hosted it on an Oracle Cloud instance
- Programmed a feature to securely save data every 30 seconds using multithreading to achieve concurrency, locks to ensure process synchronization, and file input/output to store and load data when requested.

Password Manager | Python, Django, SQLite3, Cryptography

04/2025 – 06/2025

- Programmed password manager web application using Django framework, allowing users to securely store, generate, edit and view their account credentials.
- Implemented data protection by hashing and salting master passwords and encrypting stored credentials using Cryptography library.
- Developed secure authentication system using e-mail Two-Factor Authentication (2FA), and temporary tokens.
- Protected from web attacks by utilizing Django's Object-Relational Mapping (ORM) to prevent SQL injections, token validation to block CSRF attacks, and auto-escaping to stop XSS threats.

Experience

Karizma – Freelancing Jeddah, Saudi Arabia

12/2025 - Till now

- Optimized e-commerce store project for “Karam” company and built portfolio website.

Haselhorst Automobile - Freelancing Berlin, Germany

09/2025 – Till now

- Rebuilt legacy Microsoft access vehicle management system into modern Django backend and React frontend application. Build Advanced database to store data and generate document using stored data.
- Implemented secure authorization and 2-factor authentication using one-time tokens. Enhanced user experience with pagination, internationalization, autosaving forms, and underwhelming minimal user interface.
- Utilized AI development tools to save time and detect bugs, and deployed using Docker and hosted on AWS cloud instance.

Technical & Soft Skills

- **Programming Languages:** Python, Java, JavaScript/Typescript, R, HTML, CSS, SCSS, LaTeX, Markdown.
- **Frameworks & Libraries:** React, Vue.js, Next.js, Node.js, Express.js, Nest.js, Django, Bootstrap, Tailwind CSS, NumPy, Matplotlib, Seaborn, Pandas.
- **Databases:** SQLite3, MySQL, MongoDB.
- **Tools & Cloud Environments:** Git, Jupyter Notebook, Oracle & AWS Cloud, Docker, AI development tools
- **Core Concepts:** Machine Learning, Predictive modeling, Data structures & Algorithms, Concurrency (Multithreading), Object-Oriented Programming (OOP), Cybersecurity (Firewalls, Encryption, Hashing, Salting, 2FA, Authentication, Authorization), Unit testing.
- **Soft Skills:** Adaptability to changes, Agile learning (Growth Mindset), Target-Oriented Problem Solving, and Teamwork.

Languages

- English - C1
- German – A2
- Arabic – Native